

Assignment Operators in Java

Assignment operators are used to assign values to variables in Java. The basic assignment operator (=) is used to store a value in a variable, while compound assignment operators combine a mathematical operation with assignment.

1. List of Assignment Operators

Operator	Description	Example	Equivalent To
=	Assigns the right-hand value to the left-hand variable	a = 5	N/A
+=	Adds right-hand value to the left-hand variable and assigns the result	a += 5	a = a + 5
-=	Subtracts right-hand value from the left-hand variable and assigns the result	a -= 5	a = a - 5
*=	Multiplies the left-hand variable by the right-hand value and assigns the result	a *= 5	a = a * 5
/=	Divides the left-hand variable by the right-hand value and assigns the result	a /= 5	a = a / 5
%=	Assigns the remainder of the division operation to the left-hand variable	a %= 5	a = a % 5

2. Basic Assignment Operator (=)

The simplest assignment operator is = , which assigns the value on the right-hand side to the variable on the left-hand side.

Example:

```
int a = 10; // Assigns 10 to variable a
```

3. Compound Assignment Operators

Compound operators simplify the code by combining arithmetic or bitwise operations with assignment. Instead of writing `x = x + y`, you can write `x += y`.

Example: Arithmetic Compound Operators

```
public class ArithmeticAssignmentDemo {
    public static void main(String[] args) {
        int a = 10;

        a += 5; // Equivalent to a = a + 5
        System.out.println("a += 5: " + a);

        a -= 3; // Equivalent to a = a - 3
        System.out.println("a -= 3: " + a);

        a *= 2; // Equivalent to a = a * 2
        System.out.println("a *= 2: " + a);

        a /= 4; // Equivalent to a = a / 4
        System.out.println("a /= 4: " + a);

        a %= 3; // Equivalent to a = a % 3
        System.out.println("a %= 3: " + a);
    }
}
```

4. Common Mistakes

1. Confusing `=` with `==`:

- `=` is the assignment operator, while `==` is the equality operator.

```
int a = 10; // Correct
if (a == 10) { // Checks if a equals 10
    System.out.println("Equal");
}
```

2. Using invalid types:

- The types on the left and right must be compatible.

```
int a = 10; // Valid
double b = 5.5; // Valid
```

```
// a = b; // Error: Type mismatch  
a = (int) b; // Correct with explicit cast
```

5. Comprehensive Example

```
public class AssignmentOperatorsDemo {  
    public static void main(String[] args) {  
        int a = 15, b = 5;  
  
        // Basic Assignment  
        System.out.println("a = " + a);  
  
        // Compound Assignment  
        a += b; // a = a + b  
        System.out.println("a += b: " + a);  
  
        a -= b; // a = a - b  
        System.out.println("a -= b: " + a);  
  
        a *= b; // a = a * b  
        System.out.println("a *= b: " + a);  
  
        a /= b; // a = a / b  
        System.out.println("a /= b: " + a);  
  
        a %= b; // a = a % b  
        System.out.println("a %= b: " + a);  
    }  
}
```

6. Output

```
a = 15  
a += b: 20  
a -= b: 15  
a *= b: 75  
a /= b: 15  
a %= b: 0
```